The two remaining rank-six rings: modular reconstruction and finite Morita reductions

Research supplement

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Abstract

Both $R6_{4,155}$ and $R6_{8,1660}$ remain unresolved. This supplement determines the complete candidate modular datum of the rank-26 semion centralizer forced by $R6_{4,155}$, and proves a finite Morita reduction for $R6_{8,1660}$. The modular candidate passes all exact checks stated below; it is not a construction of a fusion category. The Morita reduction excludes two kinds of module or dual-ring alternatives, and leaves three possible Morita-dual based rings. All new calculations are small exact computations; no full center enumeration is used.

1 Status and the scope of the deductions

The starting point is the completed multiplicity-nine census release of 1 October 2026. Its spherical/Galois reduction permits work in a pseudo-unitary representative for both unresolved rings. Its retained restriction vectors, low S rows and twist constraints are inputs to this supplement, with their existing certificates unchanged. The numbers of realized and excluded original rank-six rings do not change. All 8,809 rings of exact multiplicity nine remain excluded; the two unresolved rings have exact multiplicities four and eight.

Ring	New conclusion
$R6_{4,155}$	A unique complete rank-26 modular candidate, with nonnegative integral Verlinde coefficients; exactly ten self-dual objects in this factor. The original rank-52 center would have twenty self-dual objects. Three possible connected-étale underlying objects are determined.
$R6_{8,1660}$	Every indecomposable module category has six simples. Its Morita dual has rank six or eight, with faithful action. Exact Gram reconstruction and a weighted second-indicator obstruction leave three possible dual based rings.

Necessary modular data and based modules do not supply associators or algebra structures. In particular, none of the nontrivial module categories below is asserted to exist. No classification count is changed on the strength of these necessary conditions.

2 Complete modular reconstruction for $R6_{4,155}$

Write $d = 4 + 3\sqrt{2}$, $D = 72 + 48\sqrt{2}$, and $z = e^{2\pi i/24}$. The established reduction gives

$$Z(\mathcal{C}) \simeq \mathcal{S} \boxtimes \mathcal{B}, \qquad \operatorname{rank} \mathcal{B} = 26, \qquad \dim \mathcal{B} = 144d^2,$$

where $\mathcal{S}$ is the semion category. The multiplicative central charge of $\mathcal{B}$ is z^{-3}. Thus

$$s = \frac{S_{\mathcal{B}}}{12d}, \qquad t = zT_{\mathcal{B}}$$

obey $s^2 = C$ and $(st)^3 = C$, where C is charge conjugation. Dong–Lin–Ng's congruence and Galois theorem gives a representation of $\mathrm{SL}_2(\mathbb{Z}/24\mathbb{Z})$ with entries in $\mathbb{Q}(z)$; see arXiv:1201.6644, Theorem II and Section 4. All labels in this section are the zero-based labels of the frozen rank-26 restriction list; the vacuum is label 1.

2.1 Two inexpensive duality obstructions

Lemma 2.1 (Determinant obstruction)**.** *If a modular representation has $s^2 = C$ and $(st)^3 = C$, then $\det C = (\det t)^{-6}$. For the retained spectrum of $\mathcal{B}$, the number r of self-dual simples satisfies $r \equiv 26 \pmod 4$.*

Proof. Taking determinants gives $\det s = (\det t)^{-3}$. The twist exponents in $T_{\mathcal{B}}$ sum to 258, so $\det t = z^{258+26} = z^{20}$ and $\det C = 1$. An involution on 26 labels with r fixed labels has determinant $(-1)^{(26-r)/2}$. The previously allowed value $r = 12$ is therefore impossible; the two remaining values are 10 and 14. □

For every a coprime to 24, $a^2 = 1 \pmod{24}$. Hence the signed Galois matrix is

$$G_a = \sigma_a(s)s^{-1} = \rho(aI).$$

It is central, involutory, and commutes with s, t. Its simultaneous eigenspaces have rational bases. If G_a acts on such a space by a character $\chi(a)$, each entry of its s block transforms by that character. Write χ_i, χ_2, χ_3 for the characters belonging to $i, \sqrt{2}, \sqrt{3}$.

The exact character-table calculation described below forces a 12-dimensional trivial-character block, a six-dimensional χ_2 block, a six-dimensional χ_i block, and a final two-dimensional block. That last block has character χ_i when the two odd $2d$ pairs are dual pairs, and χ_3 when both pairs are self-dual.

Lemma 2.2 (Rational-square obstruction)**.** *The self-dual odd-2d branch is impossible. Consequently $\mathcal{B}$ has exactly ten self-dual simples.*

Proof. In the self-dual branch use the equal-norm rational basis $e_{12} - e_{13}, e_{14} - e_{15}$. On this χ_3 space,

$$t = \mathrm{diag}(i, iw), \qquad s = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \qquad w = e^{2\pi i/3},$$

and $a, b, c \in \sqrt{3}\,\mathbb{Q}$. The block is irreducible, so $b \neq 0$. The relation $s^2 = I$ gives $c = -a$. The off-diagonal entry of $sts = t^{-1}st^{-1}$ gives

$$a = \frac{1}{i(iw)(i - iw)} = \frac{1}{\sqrt{3}}, \qquad b^2 = \frac{2}{3}.$$

Thus $(b/\sqrt{3})^2 = 2/9$, impossible for a rational number. This is a field obstruction, not a numerical approximation. □

2.2 Exact determination of every S entry

The replay first constructs and verifies the class algebras and character tables of $\mathrm{SL}_2(\mathbb{Z}/8\mathbb{Z})$ and $\mathrm{SL}_2(\mathbb{Z}/3\mathbb{Z})$. Their 210 tensor-product irreducibles exhaust the irreducibles at level 24. There are 718 dimension-26 decompositions with the prescribed t spectrum. Signed Galois permutations and the known nonreal-character rows leave eight. The known χ_2 block has $\mathrm{Tr}(st) = 0$ and leaves four. The preceding rational-square obstruction removes the self-dual branch.

On the mandatory χ_i six-space, take the pair-difference basis

$$(2,5),\ (3,6),\ (4,7),\ (20,21),\ (16,17),\ (18,19),$$

where (a, b) denotes $e_a - e_b$. The block is

$$s = \frac{i}{2}\begin{pmatrix} 0 & 0 & 1 & -1 & -1 & -1 \\ 0 & 0 & -1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 & 1 & -1 \\ -1 & 1 & 0 & 0 & 1 & -1 \\ -1 & -1 & 1 & 1 & 0 & 0 \\ -1 & -1 & -1 & -1 & 0 & 0 \end{pmatrix}.$$

The last two rows are known low rows. Their modular relations leave three rational variables; their square relations force them to $0, 1/2, 0$. The additional χ_i two-space has $s = iI$.

Only the rational 12-dimensional block remains. Its basis is

$$e_1 - e_{23},\ e_0 - e_{22},\ e_2 + e_5,\ e_3 + e_6,\ e_4 + e_7,\ e_{20} + e_{21},$$
$$e_8,\ e_9,\ e_{10},\ e_{11},\ e_{12} + e_{13},\ e_{14} + e_{15}.$$

Let P have these basis vectors as columns, $N = P^{\mathsf{T}}P = \mathrm{diag}(2, 2, 2, 2, 2, 2, 1, 1, 1, 1, 2, 2)$, and $H = P^{\mathsf{T}}sP$. The equations from its known first two rows leave ten rational variables. Two further linear equations set one of these to zero, leaving nine variables. Expanding the remaining modular relations in the cyclotomic basis gives 187 rational polynomials of degree at most two. Row reduction on their monomials produces the nine linear consequences

$$\begin{array}{lll} h_{7,11} = -1/3, & h_{7,7} = -1/6, & h_{7,8} = 1/3, \\ h_{7,9} = 1/3, & h_{8,11} = 2/3, & h_{8,8} = -1/6, \\ h_{8,9} = -1/6, & h_{9,11} = -2/3, & h_{9,9} = -1/6. \end{array}$$

Substitution satisfies all the original equations. Thus the solution is unique; the final replay uses rational linear elimination, not a Gröbner calculation. The integer matrix $6H$ and all basis conventions are recorded in `matrix_blocks.json`. Its trace is $\mathrm{Tr}(N^{-1}H) = -2$, selecting the $3 + 9$ representation type and excluding the $3 + 3 + 3 + 3$ alternative.

Theorem 2.3 (Unique modular candidate)**.** *In the retained normalization and labels, the necessary rank-26 modular datum is unique. Its modular representation has irreducible dimensions* $9, 6, 4, 3, 2, 1, 1$. *Every one of its* 26^3 *Verlinde coefficients is a nonnegative integer, with maximum 9.*

Proof. The preceding block determinations exhaust the 26-dimensional space. For each block its contribution to the original normalized matrix is $PN^{-1}HN^{-1}P^{\mathsf{T}}$. Their sum is therefore forced. The exact replay checks every matrix entry of $s^2 = C$ and $sts = t^{-1}st^{-1}$, all known low rows, and every Verlinde coefficient in $\mathbb{Q}(z)$. A separate implementation verifies all fusion associativity identities and exact diagonalization by the reconstructed s, as well as direct Verlinde computations for four selected fusion matrices. These are checks of the forced candidate, not associators for a fusion category. □

2.3 What the completed data pass

All 456,976 ordered four-object Vafa equations hold exactly modulo 24. The ordinary second indicators agree with duality: the ten self-dual simples have indicator $+1$ except label 24, which has indicator -1. It occurs with multiplicity zero in every $U^* \otimes U$ for simple U. The 182 ordinary cyclic spectra for the 26 simples at orders $2, 3, 4, 6, 8, 12, 24$ have nonnegative integral eigenvalue multiplicities. All 676 products also agree with the frozen restriction functor to the original rank-six ring.

Using the full rank-52 matrix for $Z(\mathcal{C})$ strengthens the previous low-target generalized-indicator checks. All 7,176 cyclic spectra for 52 central targets, six original simples and orders 2 through 24 pass, together with 312 degree-one normalization identities. For $\mathcal{B}$ itself, all 17,576 generalized

second-indicator blocks and all 17,576 degree-one identities pass. The trace formula is Ng–Schauenburg, arXiv:0806.2493, Corollary 5.6, with Lemmas 2.5–2.7 and Proposition 2.8. In particular a target of twist θ requires the normalization $E^n = \theta^{-1}$ before cyclic Fourier inversion. These are finite necessary checks. The exponent 24 alone does not turn them into a proof of feasibility at every tensor order.

No known category has been identified with these data. Matching dimensions alone is insufficient, and the new near-group constructions discussed next do not provide the required condensation.

3 Categorical consequences of the rank-26 reconstruction

The results below give a construction obstruction and restrictions on algebras in the newly reconstructed fusion ring. They do not establish or exclude a categorification of $R_6^{4,155}$. For the rank-26 category $\mathcal{B}$, we use the numbering of the accompanying reconstruction: the unit is object 1, the fermion is object 0, and

$$d = 4 + 3\sqrt{2}, \qquad b = 2 + 4d = 18 + 12\sqrt{2}, \qquad A = 23, \quad L = 25,$$

where $\mathrm{FPdim}(A) = b - 1$, $\mathrm{FPdim}(L) = b$, and $\mathrm{FPdim}(\mathcal{B}) = 8b^2$. All assertions about $\mathcal{B}$ are conditional on a positive ribbon categorification of the reconstructed data.

3.1 The new order-32 near-group centers cannot give the target by dimension-eight condensation

Radchenko–Wheeler construct complex near-group categories of type $(G, |G|)$ for every product of two cyclic groups; unitarity is not asserted. Their primary source is *Real quadratic fields and finite quantum dilogarithms I*, https://arxiv.org/abs/2609.21892, version 2, Section 5.4, Theorem 12 and Corollary 6. Thus their construction supplies the three order-32 groups C_{32}, $C_2 \times C_{16}$, and $C_4 \times C_8$. The following argument applies directly to their algebraic construction; it does not import a classification of unitary centers.

A half-braiding lemma. In the strict endomorphism realization, write the simple objects as α_g and ρ. The displayed identities in the cited Section 5.4 include

$$\alpha_g\rho = \rho, \quad \rho\alpha_g = \mathrm{Ad}(U_g)\rho, \quad U_gU_h = U_{g+h}, \quad \alpha_g(U_h) = \chi(g,h)U_h,$$

where χ is a nondegenerate symmetric bicharacter. There is also a nonzero split inclusion $T : \rho \longrightarrow \rho^2$. If a central object Z has multiplicity-free pointed underlying object $\bigoplus_{g\in H} \alpha_g$, then every one of its summands is central, with self-braiding $\chi(g,g)^{-1}$.

Proof. Its half-braiding with α_h is diagonal, with scalar $a(g,h)$ on label g. Write the entry of its half-braiding with ρ from label g to label k as $M_{kg}U_k$. Naturality at $U_h : \rho \to \rho\alpha_h$ and the hexagon give

$$U_hM_{kg}U_k = a(k,h)M_{kg}U_k\alpha_g(U_h).$$

Thus $M_{kg} \neq 0$ implies $a(k,h)\chi(g,h) = 1$ for every h. Nondegeneracy of χ permits at most one nonzero entry in each row of M. Since M is invertible, it is monomial, with a permutation π of the labels H. The hexagon for ρ^2 has output label $\pi^2(g)$, whereas naturality at T compares it with a map having output label $\pi(g)$. Both maps are nonzero, being compositions of isomorphisms and split inclusions. Projection onto the tagged direct summands therefore gives $\pi^2(g) = \pi(g)$. Hence π is the identity. All half-braidings preserve each summand, and the preceding scalar identity gives $a(g,h) = \chi(g,h)^{-1}$. □

Proposition. None of these three order-32 Radchenko–Wheeler categories has a connected étale algebra of dimension eight in its center. Consequently their centers cannot produce the proposed center of $R_6^{4,155}$ by a single connected-étale condensation.

Proof. Use the established two-dimension reduction to pass to a positive spherical Galois representative; a hypothetical algebra transports under the same conjugation. Put

$$r = 16 + 12\sqrt{2} = 4d, \qquad \Delta = 32 + r^2 = 8(72 + 48\sqrt{2}).$$

The formal-codegree decomposition of $I(1)$ has constituent dimensions 1, $r+1$, and 31 copies of $r+2$. Indeed the 31 nontrivial group characters have formal codegree 32, while the two remaining characters send ρ to its two quadratic roots. In particular every nonunit constituent of $I(1)$ has dimension greater than eight.

Suppose $E \in Z(\mathcal{C})$ is connected étale of dimension eight. Its underlying object is pointed, since $r > 8$. Adjunction and the preceding constituent bound give

$$\dim \mathrm{Hom}_{\mathcal{C}}(1, F(E)) = \dim \mathrm{Hom}_{Z(\mathcal{C})}(I(1), E) = 1.$$

Thus $F(E)$ is a connected separable algebra in the pointed subcategory. It is simple as a right module over itself, since its endomorphism space is $\mathrm{Hom}_{\mathcal{C}}(1, F(E))$. Each $\alpha_g \otimes F(E)$ is also simple. Free-module adjunction now shows

$$[F(E) : \alpha_g] = \dim \mathrm{Hom}_{F(E)}(\alpha_g \otimes F(E), F(E)) \in \{0, 1\}.$$

Its support is the stabilizer subgroup H of this simple module and has order eight. The lemma splits E into central invertibles. Twist-freeness of a connected étale algebra then forces $\chi(h, h) = 1$ for every $h \in H$.

For completeness the required bicharacter count is elementary and uniform over all nondegenerate choices. On C_{32}, $\chi(x, x) = \exp(2\pi i c x^2/32)$ with c odd, so the bosons are exactly the four multiples of eight. On $C_2 \times C_{16}$, write diagonal parameters $p/2, q/16$ and off-diagonal parameter $r_0/2$. Nondegeneracy forces $p = 1$ and q odd: otherwise the order-16 generator maps to a character of smaller order, or the order-two generator is in the pairing kernel or has the same character as eight times that generator. The boson equation is $8a + qb_0^2 = 0 \pmod{16}$; it forces $a = 0$ and $4 \mid b_0$. On $C_4 \times C_8$, write parameters $p/4, q/8, r_0/4$. Here p, q must be odd: if q is even the order-eight character has smaller order, while if p is even the nonzero element $(2, 4r_0)$ is in the kernel. The boson equation

$$2pa^2 + qb_0^2 + 4r_0 ab_0 = 0 \pmod 8$$

first forces b_0 even, then a even, and finally $4 \mid b_0$. There are again only four bosons. No required subgroup H exists. Finally, the source-center dimension is $\Delta^2 = 64 \dim(\mathcal{C}_{155})^2$; the local-module dimension formula forces a condensation to the target center to use an algebra of dimension eight. □

This excludes all dimension-eight connected étale algebras in these centers, not merely pointed candidates. It does not exclude other source categories, products followed by condensation, or the unresolved near-group type $(8, 16)$.

3.2 The only possible connected étale underlying objects in $\mathcal{B}$

Proposition. Every connected étale algebra in a positive ribbon categorification of the reconstructed rank-26 data has underlying object

$$1, \qquad 1 + A, \qquad 1 + A + L.$$

This statement does not assert uniqueness of algebra structures.

Proof. The twist-one simples are precisely $1, 2, 5, 23, 25$, with $2^* = 5$. The reconstructed fusion rules give

$$2 \otimes 2 = 9 + 10 + 16 + 19, \qquad 2 \otimes 5 = 1 + 8 + 11 + 22.$$

The first product has no twist-one constituent; the second has only the unit among such constituents. A connected étale algebra has a nondegenerate Frobenius pairing. If it contained object 2, it would contain 5, and multiplication of suitable copies of 2 and 5 to the unit would be nonzero. Its multiplication on $2 \otimes 2$ would vanish. Associativity on $2 \otimes 2 \otimes 5$ is then impossible: multiplying the first two factors gives zero, whereas contracting the last two to the unit and multiplying by the first gives a nonzero map. This argument also works with arbitrary multiplicities, by choosing a nonzero entry of the pairing on the multiplicity spaces. Thus neither 2 nor 5 can occur.

Write the remaining candidate as $E = 1 + sA + tL$, with $s, t \geq 0$. Its dimension is $1 - s + (s+t)b$, at most $\sqrt{\mathrm{FPdim}(\mathcal{B})} = 2\sqrt{2}\,b$. If $s + t \geq 3$, its dimension is at least $3b - 2$, which exceeds this bound by four. Hence $s + t \leq 2$. Apart from the three stated objects, the possibilities have dimensions $b + 1$, $2b - 1$, and $2b + 1$, with norms

$$N(b+1) = 73, \qquad N(2b-1) = 73, \qquad N(2b+1) = 217.$$

But

$$N(\mathrm{FPdim}(\mathcal{B})) = N(8b^2) = 82944 = 2^{10}3^4.$$

The number $\mathrm{FPdim}(\mathcal{B})/\,\mathrm{FPdim}(E)^2$ must be an algebraic integer, being the dimension of the local-module category. Its norm would require 73^2 or 217^2 to divide 82944, which is false. □

The standard algebra facts used here—the Frobenius pairing, twist-freeness in the positive setting, and local-module dimension—are discussed in Davydov–Müger–Nikshych–Ostrik, *The Witt group of non-degenerate braided fusion categories*, Section 3, https://arxiv.org/abs/1009.2117.

Fusion subcategories and restriction. There are exactly four fusion subcategories: Vec, sVec, the rank-16 fermion centralizer, and $\mathcal{B}$. This conclusion follows directly from a few support properties of the full fusion table. Every noninvertible simple other than $4, 7$ has 22 in its product with its dual. For $4, 7$, that product contains 8, whose product with its dual contains 22. The square of 22 contains every even simple except the fermion; it contains 23, and $22 \otimes 23$ contains the fermion as well. Thus every noninvertible even simple generates the entire even category. A subcategory containing it and an odd object is faithfully C_2-graded with that even component. Equality of the dimensions of grading components forces it to be all of $\mathcal{B}$. The accompanying integer checker also verifies every one of the 676 restriction-product identities

$$W(i)W(j) = \sum_k N_{ij}^k W(k)$$

in the original rank-six ring. Thus the full reconstructed fusion rules pass the restriction-functor consistency check, not only its previously available low products. The checks and exact support sets are recorded in `full_fusion_categorical_receipt.json`.

3.3 The canonical intermediate condensation adds no new near-group category

When $\mathcal{B}$ comes from a categorification $\mathcal{C}$ of the rank-six ring, the restricted adjoint algebra is $E = 1 + A + L$. Its even subalgebra $E_0 = 1 + A$ has dimension b. The local category $\mathcal{K} = \mathcal{B}^0_{E_0}$ has dimension eight. The summand L, as an E_0-module, is an invertible boson ℓ; the free fermion $f_0 = fE_0$ is distinct, and $f_0\ell \neq \ell$ has the same underlying object L. Their twists $1, -1, 1, 1$ give a toric-code factor. Preservation of central charge then gives

$$\mathcal{K} \simeq \mathrm{ToricCode} \boxtimes \mathrm{Semion}^{\mathrm{rev}}.$$

In fact the category of all modules $\mathcal{D} = \mathcal{B}_{E_0}$ is tensor equivalent to $\mathcal{C} \boxtimes \mathrm{Rep}(C_2)$. Here are details preventing an inference from merely fixing simple objects. The boson ℓ realizes $\mathcal{D}$ as a C_2-equivariantization of $\mathcal{C}$. Its action fixes X, Y, since each is the free image of a d-dimensional simple of $\mathcal{B}$. It fixes every invertible as well, since their unique central lifts have distinct twists $1, -1, i, -i$. Consequently $\mathcal{D}$ has eight invertibles and four simples of dimension d.

The local category $\mathcal{K}$ supplies all eight invertibles, while $Z(\mathcal{D}) \simeq \mathcal{B} \boxtimes \mathcal{K}^{\mathrm{rev}}$ has sixteen. The kernel of the central forgetful map, and hence the character group of the universal grading group U, has order two. Equal dimensions of grading components, by comparing coefficients in $a + cd^2 = (16 + 32d)/|U|$, imply $c = 4/|U|$. Thus $|U|$ divides four; such a group is abelian, and therefore $U = C_2$. The adjoint component has four invertibles and two noninvertibles. Its two noninvertibles must map respectively to X, Y under de-equivariantization: two mapping to X would contradict the occurrence of Y in X^2, and conversely. Their products force four distinct invertible images. Hence $\mathcal{D}_{\mathrm{ad}} \to \mathcal{C}$ is a tensor equivalence. The central boson ℓ lies outside this component and gives the asserted tensor-product decomposition. This canonical construction therefore supplies no exchange action producing a near-group category of type $(8, 16)$.

4 Finite Morita reduction for $R6_{8,1660}$

Proposition 4.1 (A general formal-codegree test for modules)**.** *Let $\mathcal{F}$ be pseudo-unitary with commutative Grothendieck ring. For an indecomposable module category, let m_χ be the multiplicity of the character χ in its complexified based module, and let f_χ be the formal codegree. Then*

$$\sum_{\chi} \frac{m_\chi}{f_\chi} \leq 1.$$

If equality holds, the Morita dual has rank $\sum_\chi m_\chi^2$, and its action is faithful and equals the full commutant. In this case its simple action matrices must give a square nonnegative integral Gram factorization of the linking-category matrix below. Failure of that bounded factorization is an obstruction to the based module being a module category.

Proof. Ostrik's module-character reciprocity identifies m_χ with the multiplicity in the module's Lagrangian algebra of the central simple of dimension $\dim(\mathcal{F})/f_\chi$. Positivity and the dimension of that algebra give the inequality. Equality exhausts its support inside the canonical algebra. Its endomorphism algebra is consequently $\bigoplus_\chi M_{m_\chi}(\mathbb{C})$ and acts faithfully on the module Hom space. The next proof details these identifications and their precise primary source. The linking associativity identity then gives the stated Gram factorization. □

This note gives necessary module-category conditions, not an existence or nonexistence theorem. No braiding on $\mathcal{C}$ or its relative center is assumed. Let $\mathcal{C}$ have simples $1, g, X, gX, \sigma, T = \sigma X$ and rules

$$g^2 = 1, \quad g\sigma = \sigma, \quad \sigma^2 = 1 + g, \qquad X^2 = 1 + (3(1 + g) + 4\sigma)X.$$

Put $d = 3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$, so $\mathrm{FPdim}\,\mathcal{C} = 4(1 + d^2)$. All matrices below use the action on the basis of simple objects of a finite semisimple module category.

Lemma 4.2. *Every indecomposable based module over the Ising fusion ring, whose simple action matrices obey duality transpose, is its three-object regular module. Consequently every $\mathcal{C}$-module category restricts to a direct sum of regular Ising module categories at the level of based modules.*

Proof. Write $H = N_\sigma$ and $G = N_g$. Then G is a permutation involution, $H = H^{\mathsf{T}} \geq 0$, $GH = H$, and $H^2 = I + G$. An indecomposable component of H has spectral radius $\sqrt{2}$: this follows either from its dimension vector or directly from these identities and $GH = H$. There are no entries of H greater than one. There are no loops: a loop connected to any different vertex has norm at least the golden ratio, while an isolated loop contradicts $H^2 = I + G$. No vertex has degree three or more, and a path with four vertices already has norm greater than $\sqrt{2}$. Thus every connected component is the three-vertex path A_3; the possibilities of one or two vertices contradict $H^2 = I + G$. Its two leaves are exchanged by G and its middle vertex is fixed. These are the regular Ising matrices. □

Theorem 4.3 (Uniform module rank)**.** *Every indecomposable finite semisimple $\mathcal{C}$-module category has exactly six simple objects. Its restriction to $\mathcal{I} = \langle g, \sigma \rangle$ consists of two regular Ising blocks.*

Proof. Initially suppose there are n blocks. Order each block as $1, g, \sigma$. The commutant of the two regular Ising matrices consists exactly of

$$J(a,b,c) = \begin{pmatrix} a & b & c \\ b & a & c \\ c & c & a+b \end{pmatrix}.$$

Since X commutes with g, σ, and is self-dual, its action has form $N_X = P \otimes I + Q \otimes N_g + R \otimes N_\sigma$, where P, Q, R are symmetric nonnegative integral n by n matrices. Set $U = P + Q$, $V = P - Q$. The X^2 relation gives

$$V^2 = I, \quad U^2 + 2R^2 = I + 6U + 8R, \quad UR + RU = 6R + 4U. \tag{1}$$

The matrix $M = U + \sqrt{2}R$ therefore satisfies

$$M^2 - (6 + 4\sqrt{2})M - I = 0. \tag{2}$$

It is irreducible: otherwise a union of Ising blocks would be invariant under all the generators of $K_0(\mathcal{C})$, contradicting indecomposability. Hence its positive Perron eigenvalue d is simple. The polynomial in (2) is irreducible over $\mathbb{Q}(\sqrt{2})$: its discriminant is $72 + 48\sqrt{2} = (4\sqrt{3} + 2\sqrt{6})^2$, whose square root does not belong to $\mathbb{Q}(\sqrt{2})$. Thus the characteristic polynomial of M is a power of (2); simplicity of the Perron root forces that power to be one. Therefore $n = 2$. □

Theorem 4.4 (Complete based-module list)**.** *Up to interchanging the two Ising blocks, the possible matrices U, R are exactly*

$$U = \begin{pmatrix} a & b \\ b & 6-a \end{pmatrix}, \qquad R = \begin{pmatrix} c & f \\ f & 4-c \end{pmatrix},$$

with the following four rows. These are based-module possibilities; their associativity constraints have not been solved.

type	a	b	c	f	$V = P - Q$
I	*0*	*1*	*0*	*0*	*signed transposition*
II	*2*	*3*	*2*	*2*	*signed transposition*
III	*1*	*2*	*0*	*1*	$\mathrm{diag}(\epsilon_1, \epsilon_2)$
IV	*1*	*2*	*1*	*2*	$\mathrm{diag}(\epsilon_1, \epsilon_2)$

Here $\epsilon_i \in \{1, -1\}$ and $P = (U + V)/2$, $Q = (U - V)/2$. The two signed transpositions give equivalent modules by swapping the two leaves in one Ising block. Thus there are ten labelled-X based modules, or six after additionally allowing the ring automorphism $X \leftrightarrow gX$.

Proof. Equation (2) and $n = 2$ give $\operatorname{tr} U = 6$ and $\operatorname{tr} R = 4$. The matrix V is symmetric integral and $V^2 = I$, so it is a signed permutation. It must satisfy $V \equiv U \pmod 2$, with $U \pm V \geq 0$ entrywise. Put $w = a - 3$, $s = c - 2$. The two equations (1) become

$$ws + bf = 6, \qquad w^2 + b^2 + 2(s^2 + f^2) = 18. \tag{3}$$

All variables are integral; $|w| \leq 3$, $|s| \leq 2$, and $b, f \geq 0$. Cauchy–Schwarz implies $3 \leq s^2 + f^2 \leq 6$. The only sums of two integral squares in this interval are 4 and 5. For value 4, the squared norm of (w, b) is 10; for value 5, it is 8. Substitution in $ws + bf = 6$ gives exactly the four rows and their block exchanges. Parity gives precisely the displayed options for V. Conversely these matrices are nonnegative integral and satisfy every fusion relation and transpose relation. Their X action connects the two Ising blocks, so they are indecomposable based modules. □

Corollary 4.5 (Four small algebra supports)**.** *Every indecomposable $\mathcal{C}$-module category has a connected separable internal-end algebra on one of the following underlying objects:*

$$1, \qquad 1+X+gX+2T, \qquad 1+X \text{ or } 1+gX, \qquad 1+X+T \text{ or } 1+gX+T. \tag{4}$$

Consequently these supports exhaust the possible Morita classes, with no assertion that the nontrivial supports admit algebra structures or that different structures give different Morita classes.

Proof. Choose a leaf m in the first Ising block of each row. The coefficient of a simple Z in $\underline{\mathrm{End}}(m)$ is $(N_Z)_{mm}$. Thus this internal end is $1+P_{11}X+Q_{11}gX+R_{11}T$. Substitution gives (4). The standard internal-Hom construction supplies a connected separable algebra and realizes the original module category as its modules. The first row has internal end 1 and is necessarily the regular module category, even categorically. □

The Perron dimension ratio of the second Ising block to the first is, respectively,

$$d, \quad 3-2\sqrt{2}+2\sqrt{3}-\sqrt{6}, \quad \sqrt{2}+\sqrt{3}, \quad (\sqrt{2}+\sqrt{6})/2.$$

The dimension quotients $\mathrm{FPdim}\,\mathcal{C}/\,\mathrm{FPdim}\,Q$ for the three nontrivial supports in (4) are, respectively,

$$144-96\sqrt{2}+80\sqrt{3}-56\sqrt{6}, \qquad 24+8\sqrt{6}, \qquad 12+4\sqrt{3}.$$

All are algebraic integers, so elementary dimension divisibility does not exclude these targets. The two choices of the second diagonal sign in types III and IV are not removed by the chosen algebra support. They may encode different algebra/module associativity structures.

Theorem 4.6 (Full-center saturation and ranks of all Morita duals)**.** *Use the established reduction to the pseudo-unitary form of $\mathcal{C}$. For any indecomposable module category $\mathcal{M}$, the dual fusion category $\mathcal{D}=\mathrm{Fun}_{\mathcal{C}}(\mathcal{M},\mathcal{M})$ has rank six or eight. More precisely,*

matrix V	$\mathrm{rank}\,\mathcal{D}$	$K_0(\mathcal{D})\otimes\mathbb{C}$
$\mathrm{spec}(V)=\{1,-1\}$	6	$\mathbb{C}^6$
$V=I$ *or* $-I$	8	$\mathbb{C}^4\oplus M_2(\mathbb{C})$.

The action of $K_0(\mathcal{D})\otimes\mathbb{C}$ on $K_0(\mathcal{M})\otimes\mathbb{C}$ is faithful and is the full commutant of the $K_0(\mathcal{C})$ action.

Proof. Write $D=\mathrm{FPdim}\,\mathcal{C}$ and let A_χ denote the six simple summands of the canonical algebra $I_{\mathcal{C}}(1)$. They occur once, because $K_0(\mathcal{C})$ is commutative. Ostrik's Theorem 2.13 gives $\dim A_\chi = D/f_\chi$. Let $L_{\mathcal{M}}$ be the Lagrangian algebra in $Z(\mathcal{C})$ corresponding to $\mathcal{M}$; under the Morita equivalence of centers it is $I_{\mathcal{D}}(1)$. The precise module-character reciprocity needed here is given by Ostrik's Remark 2.19: the bimodule $K_0(\mathcal{M})\otimes\mathbb{C}$ is identified with

$$\mathrm{Hom}_{Z(\mathcal{C})}(I_{\mathcal{C}}(1),L_{\mathcal{M}})$$

over the corner algebras $\mathrm{End}(I_{\mathcal{C}}(1))$ and $\mathrm{End}(L_{\mathcal{M}})$. This follows by applying that remark to the linking multi-fusion category of $\mathcal{C}$ and $\mathcal{M}$. Its twist hypothesis is automatic in the pseudo-unitary case, as stated at the end of the same remark. Thus the multiplicity of χ in the module character is exactly $[L_{\mathcal{M}}:A_\chi]$.

The four characters with $g=1$ each occur once: each of the two Ising spin evaluations gives a two-dimensional matrix with its two distinct roots. The two characters with $g=-1$, $\sigma=0$, $X=\pm1$ occur with multiplicities m_+,m_- equal to the eigenvalue multiplicities of V; hence $m_++m_-=2$. Both have formal codegree 4. Consequently the dimensions of all the already specified constituents of $L_{\mathcal{M}}$ sum to exactly D, just as those of $I_{\mathcal{C}}(1)$ do. Since $\dim L_{\mathcal{M}}=D$ and all dimensions are positive, there are no additional constituents. Therefore

$$L_{\mathcal{M}}=\bigoplus_{\chi:g=1}A_\chi \;\oplus\; m_+A_+\oplus m_-A_-, \qquad \dim\mathrm{End}(L_{\mathcal{M}})=4+m_+^2+m_-^2.$$

Ostrik's Corollary 2.16 identifies this endomorphism algebra with the complexified Grothendieck algebra of $\mathcal{D}$, proving the table. Every simple constituent of $L_{\mathcal{M}}$ also occurs in $I_{\mathcal{C}}(1)$. Consequently the action of $\mathrm{End}(L_{\mathcal{M}})$ on the displayed Hom space sees every matrix block and is faithful. Its block decomposition also shows that it is the full commutant. □

A first forced Morita-dual ring. For type III with $V = \mathrm{diag}(1, -1)$, the rank-six dual, if the module category exists, has the original fusion ring $R6_{8,1660}$. Here is a short exact reconstruction. Write N_x for the six known action matrices on $\mathcal{M}$, and Q_y for the unknown right action matrices of the dual. Associativity in the linking multi-fusion category and Frobenius reciprocity imply

$$\sum_x N_x[a,b]N_x[c,d] = \sum_y Q_y[c,a]Q_y[d,b]. \tag{5}$$

The central fermion of $\mathcal{C}$ gives a dual invertible whose action is N_g, so the two matrices I, N_g are fixed. Choose a leaf a in the first Ising block. The diagonal entry of (5) at the coordinate (a,a) is 2; after removing I, precisely one further simple matrix Q_Y has entry 1 there. It is therefore the corresponding Gram column minus the column of $I \otimes I$. Its g-translate is another simple. After removing these four matrices, the residual Gram matrix has rank two. At coordinate $(a, \sigma a)$ its diagonal is 2, so both remaining matrices have entry 1. That Gram column is their sum v; if H is the residual Gram matrix, $2H - vv^{\mathsf{T}}$ is the outer product of their difference. This determines them up to exchange. The resulting matrices, called Q_Z, Q_s, satisfy

$$s^2 = 1 + g, \qquad sY = Z, \qquad Y^2 = 1 + 3Y + 3gY + 4Z.$$

The exact script `reconstruct_type_iii.py` verifies (5), nonnegative integrality, linear independence, and all fusion products. Thus this branch is a possible Morita self-partner, not an exclusion.

Theorem 4.7 (Complete integral Morita Gram reduction)**.** *The two type-IV based modules with $V = I$ or $V = -I$ cannot be module categories. For the other types, equation (5), the central fermion, and the based-ring axioms leave exactly the following action bases. Set*

$$a_0 = 2 + \sqrt{2} + \sqrt{3} + \sqrt{6}, \quad b_0 = 2 + 2\sqrt{2} + \sqrt{3} + \sqrt{6}, \quad c_0 = 5 + 4\sqrt{2} + 3\sqrt{3} + 2\sqrt{6}.$$

module type	*number*	*dimensions of forced dual rings*
II	*1*	$1, 1, \sqrt{2}, d, d, \sqrt{2}d$
III, mixed signs	*1*	$1, 1, \sqrt{2}, d, d, \sqrt{2}d$
III, equal signs	*2*	$1, 1, d, d, a_0, a_0, b_0, b_0$
IV, mixed signs	*2*	$1, 1, \sqrt{2}, d, d, \sqrt{2}d$ *or* $1, 1, 1 + \sqrt{3}, a_0, a_0, c_0$
IV, equal signs	*0*	*none.*

The three occurrences of the first dimension list have the original ring. The second rank-six ring is self-dual with commuting generators g, S, Y satisfying

$$g^2 = 1, \quad gS = S, \quad S^2 = 1 + g + 2S, \qquad Y^2 = 1 + (1 + g + 2S)Y.$$

Its maximal structure constant is 16*. The two rank-eight rings have maximal structure constant* 3 *and are not isomorphic: in one the b_0 pair is a dual pair and all other simples are self-dual; in the other the a_0 pair is a dual pair and all other simples are self-dual.*

Proof. Here is a complete finite calculation with explicit bounds. It is implemented in `reconstruct_dual_actions.py`; its output is `dual_actions_receipt.json`. Flatten action matrices to vectors. Equation (5) gives a known nonnegative symmetric 36 by 36 matrix $K = \sum_x N_x \otimes N_x$. Subtract the outer products of the known matrices I, N_g. For type III also subtract the two matrices forced in the preceding reconstruction. The residual matrix H has rank r, equal to the number of remaining dual simples.

Choose r coordinates whose principal submatrix B is invertible. Any remaining simple column q is determined by its coordinates v there: $q = H_{*,J}B^{-1}v$. Because there are exactly r remaining columns, their square factorization of B implies

$$v^{\mathsf{T}}B^{-1}v = 1, \qquad 0 \le v_i \le \lfloor \sqrt{B_{ii}} \rfloor.$$

Conversely, an orthogonal basis of r such vectors, lifting to nonnegative integral columns, gives the whole Gram factorization. This is a bounded integer ellipsoid enumeration, followed by a clique search for orthogonal bases. The numbers of candidate columns and orthogonal bases for rows II, III mixed, III equal, IV mixed, IV equal are respectively

$$(4,1), \quad (2,1), \quad (10,2), \quad (12,3), \quad (12,0).$$

Testing nonnegative integral products, closure under transpose, and the unit coefficient duality rule leaves respectively $1,1,2,2,0$ fusion action bases. Faithfulness proved above makes their multiplication the actual dual fusion ring. The receipt records every resulting matrix and every structure constant, so the counts and identifications require no numerical optimizer. The signs $V \mapsto -V$ amount to relabelling X and gX, which leaves $K = \sum_x N_x \otimes N_x$ unchanged. □

Corollary 4.8 (A Morita second-indicator exclusion)**.** *The rank-eight ring in which the a_0 pair is nonself-dual cannot occur as a Morita dual of $\mathcal{C}$. In the other rank-eight possibility every self-dual simple has second indicator $+1$.*

Proof. For a pseudo-unitary category $\mathcal{F}$, put $W_2(\mathcal{F}) = \sum_Z \nu_2(Z) \dim Z$. The center indicator formula gives $\tau_2(Z(\mathcal{F})) = \dim(\mathcal{F}) W_2(\mathcal{F})$. Positive spherical structures and the twist are preserved by Morita equivalence of centers, so W_2 is a Morita invariant here. The established indicators of $\mathcal{C}$ give

$$W_2(\mathcal{C}) = 2 - \sqrt{2} + (2+\sqrt{2})d = 12 + 6\sqrt{2} + 6\sqrt{3} + 4\sqrt{6}.$$

If the a_0 pair is nonself-dual, the possible indicator sum has form $u + s_d d + s_b b_0$, where $u \in \{0,2\}$ and $s_d, s_b \in \{-2,0,2\}$. Comparing $\sqrt{3}$ and $\sqrt{6}$ coefficients forces $s_d = s_b = 2$. The resulting $\sqrt{2}$ coefficient is 8, not 6, a contradiction. This argument does not need a prior choice of the indicator of the nontrivial invertible. If instead the b_0 pair is nonself-dual, the same coefficient comparison gives $s_d = s_a = 2$ and then $u = 2$. Indeed $W_2(\mathcal{C}) = 2 + 2d + 2a_0$. □

Corollary 4.9 (No ordinary generalized-near-group Morita model)**.** *No category with this ring can be Morita equivalent to a generalized near-group category in which invertibles act transitively on all noninvertible simples. The same holds after any finite chain of finite group extensions, equivariantizations and inverse such operations starting from an ordinary generalized-near-group category.*

Proof. For a generalized-near-group category the noninvertibles have common dimension z, and $z^2 = k + mz$ for integers $k > 0, m \geq 0$. Its global dimension lies in $\mathbb{Q}(z)$, of degree at most two. Morita equivalence preserves FP dimension; extensions and equivariantizations multiply it by positive integers, and their inverses divide by integers. In contrast $\mathrm{FPdim}\,\mathcal{C} = 144 + 96\sqrt{2} + 80\sqrt{3} + 56\sqrt{6}$ has degree four: none of the three nontrivial automorphisms of $\mathbb{Q}(\sqrt{2},\sqrt{3})$ fixes it. □

Remaining gap. The displayed nimreps do not supply their module associators. The full-center and Gram arguments now leave only three possible Morita-dual based rings: the original rank-six ring, the displayed rank-six $\mathrm{Ad}(E_6)$ extension, and one rank-eight ring. Their existence, and the associators of the candidate modules, remain unproved. The present note excludes module/dual alternatives but does not exclude or construct $R6_{8,1660}$.

Background sources. The internal-Hom/algebra description of semisimple module categories is standard; see V. Ostrik, *Module categories, weak Hopf algebras and modular invariants*, arXiv:math/0111139. For module traces in the pseudo-unitary case and their symmetric Frobenius algebras, see G. Schaumann, *Traces on module categories over fusion categories*, especially Section 5 and Section 6, arXiv:1206.5716. The matrix reduction and its four-row solution above are proved here and do not require the module-trace theorem. The full-center argument uses V.

Ostrik, *Pivotal fusion categories of rank 3*, Theorem 2.13, Corollary 2.16 and especially Remark 2.19, arXiv:1309.4822, pp. 6–8. The center indicator formula is from S.-H. Ng and P. Schauenburg, *Frobenius–Schur indicators and exponents of spherical categories*, Theorem 4.1, arXiv:math/0601012.

4.1 A signed Galois–fermion cut for R6_8_1660

The following necessary condition removes one further labelled profile from the frozen 1670-profile frontier. It leaves 1669 profiles, in branches of sizes $163, 753, 753$; it does not resolve categorifiability.

Work in the established pseudo-unitary Galois form, with $\mathcal{M} = Z(\mathcal{C})$, $T = \sigma X$, and restriction vectors in the order $(1, g, X, gX, \sigma, T)$. Let h be the central fermion and let $\widehat{\alpha} = Z(\alpha)$ be the previously constructed involution. For $a \in (\mathbb{Z}/48\mathbb{Z})^{\times}$, write π_a for the Galois permutation and $\delta_a(Z) \in \{\pm 1\}$ for its sign, so that for the normalized modular matrix S,

$$\sigma_a(S_{L,Z}) = \delta_a(Z) S_{L,\pi_a Z}, \qquad \theta_{\pi_a Z} = \theta_Z^{a^2}.$$

Lemma 4.10 (The signed restriction action)**.** *For every central simple Z with restriction vector v,*

$$v_{\pi_5 Z} = \delta_5(Z) B_5 v, \qquad B_5 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 2 & 2 & -1 & -2 & -2 & 2 \\ 2 & 2 & -2 & -1 & -2 & 2 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 2 & 2 & -2 & -2 & -4 & 3 \end{pmatrix}.$$

The sign is the one making the vector nonnegative. Moreover π_5 commutes with tensoring by h and with $\widehat{\alpha}$, and preserves monodromy charge with h.

Proof. The six constituents L_χ of $I(1)$ correspond to the six characters of $K_0(\mathcal{C})$, with formal codegrees f_χ. The known induction rows are

$$S_{L_\chi,Z} = \frac{\chi(FZ)}{f_\chi}.$$

On $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, σ_5 negates both square roots. For all six characters and all six basis elements, direct arithmetic gives

$$\sigma_5\left(\frac{\chi(v)}{f_\chi}\right) = \frac{\chi(B_5 v)}{f_\chi}.$$

The character table is invertible, proving the asserted restriction formula. The verifier checks these 36 identities exactly in the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$, and checks $B_5^2 = 1$.

Write ϵ_Z for the charge. Since $S_{h,Z}/S_{1,Z} = \epsilon_Z \in \{\pm 1\}$, the column Galois identity shows that π_5 preserves charge. Also $S_{hY,Z} = \epsilon_Z S_{Y,Z}$ and $d_{hY} = d_Y$ imply $\delta_5(hY) = \delta_5(Y)$; applying Galois to the row identity and using invertibility of S gives $\pi_5(hY) = h\pi_5(Y)$. Finally, the permutation matrix of a braided autoequivalence commutes with S and is rational, so it commutes with the signed Galois matrix $S^{-1}\sigma_5(S)$. Preservation of dimensions also preserves its signs, proving commutation with $\widehat{\alpha}$. □

Lemma 4.11 (Four-divisibility of the spin fixed locus)**.** *In a restriction fiber*

$$v = (0, 0, b, c, 1, 3), \qquad b + c = 4,$$

the number of h-fixed simples is divisible by four.

Proof. The signed formula gives $B_5 v = -(0,0,c,b,1,3)$, so $\gamma = \pi_5\widehat{\alpha}$ preserves the fiber. Its simples have primitive sixteenth-root twists; therefore $\theta_{\gamma Z} = \theta_Z^{25} = -\theta_Z$. The preceding lemma implies that γ preserves the h-fixed locus. Duality preserves that locus, the restriction vector and the twist. It acts freely: a self-dual simple with this restriction would have to have indicator -1 from the odd σ multiplicity and indicator $+1$ from the odd T multiplicity. Thus each twist occurs an even number of times in the fixed locus. The bijection γ gives equal multiplicities for the two twists $\theta, -\theta$, proving divisibility by four. No claim that γ is a tensor functor is used. □

Here is the complete exclusion of the zero-based raw profile 133 in low frame zero, character labelling $(0,5,4,1,3,2)$. Remove the four forced even low/high pairs. The remaining 18 high simples have the following restrictions:

symbol	restriction	multiplicity
A	T	2
A'	$2X + 2gX + 3T$	2
B	$X + gX + 2T$	2
C	$4X + 4gX + 4T$	2
Σ	$2X + 2gX + \sigma + 3T$	8
P	$2X + 2gX + 2T$	1
Q	$4X + 4gX + 8T$	1

All these fibers are h-invariant. Let e_U, o_U, t_U be their numbers of even h-pairs, odd h-pairs and odd h-fixed simples. The signed Galois map exchanges A, A', so all three orbit counts agree for this pair. Put $a = e_A = e_{A'}$, $b = e_B$, $c = e_C$, and $z = e_\Sigma$. Then $a, b, c \in \{0,1\}$, $e_P = e_Q = 0$, and the previously proved duality parity gives $z \in \{0,2,4\}$. The even-dimension equation is exactly

$$3a + b + 8c + 3z = 15.$$

Its only solutions are

$$(a,b,c,z) = (1,0,0,4), \qquad (0,1,1,2).$$

For total remaining orbit counts E, O, H, the already established rank identity in this low frame is $O = 18 - 3E - 3$. The first solution gives $E = 6$ and $O = -3$, impossible. The second gives $E = 4$ and $O = 3$. The fibers B, C are entirely even, and P, Q have multiplicity one. Consequently

$$2o_A + o_\Sigma = 3, \qquad 0 \le o_A \le 1, \quad 0 \le o_\Sigma \le 2.$$

Thus $o_A = o_\Sigma = 1$, and $t_\Sigma = 8 - 2e_\Sigma - 2o_\Sigma = 2$, contradicting the lemma.

The portable verifier `verify.py` replays the frozen frontier, checks the signed identities, and supplies directly checked orbit-count witnesses for the other 1669 profiles. It also independently checks this two-case proof for profile 133. The broader replay imposes Galois equality of all three h-orbit counts, spin-fiber and spin-fixed four-divisibility, and the prior dimension, rank, involution and AMV/sixteenth-indicator conditions. Its receipt is `certificate.json`; no optimizer or full center fusion reconstruction is used. Passing this necessary test does not assert existence of spins, modular data, or a categorification.

5 Reproduction and remaining mathematical tasks

The accompanying archive contains the exact matrices, fusion tensors, proof notes, independent audits, and a replay runner. It is an overlay for the frozen multiplicity-nine package, whose archive SHA-256 is

`6cbed9b202f6fde6e789cbc4d0c2f00fce6`
`af4dc57072226f612fa582c1bbbbd.`

Set `RANK6_PACKAGE` to its extracted package directory and run `python replay.py` from the overlay. Arithmetic uses integers, rationals and exact cyclotomic fields. The finite scopes and receipts are listed separately, so a passing indicator check is never used as a categorification assertion.

For $R6_{4,155}$ the unresolved step is now categorical: realize the explicit modular candidate with the required algebra, or derive an obstruction beyond the checks recorded here. For $R6_{8,1660}$ the new finite targets are the algebra structures and module associators underlying the three surviving Morita-dual rings. Neither the regular module nor any of the necessary matrix solutions forces the existence of a nontrivial one. The signed-Galois fixed-locus criterion removes profile 133 in the first center branch, reducing the 1,670 previously retained profiles to 1,669, with branch counts $163, 753, 753$.